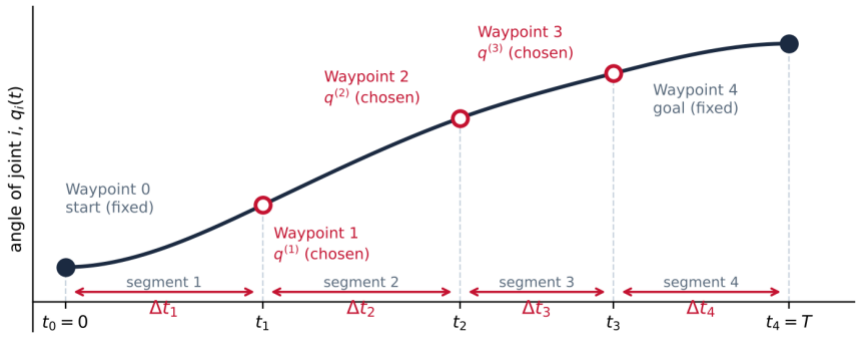
Problem Formulation



Decision variables: $\mathbf{x} = \left[q_i^{(k)},\ \Delta t_k \right] \in \mathbb{R}^{13}$, the angle of joint $i = 1,2,3$ at intermediate waypoint $k = 1,2,3$ (9) and the duration of segment $k = 1, \ldots, 4$ (4). Each segment is a cubic, $q_{ik}(\tau) = a_{ik} + b_{ik}\tau + c_{ik}\tau^2 + d_{ik}\tau^3$, so the jerk is $j_{ik} = 6d_{ik}$.

Objective

$$\min_{\mathbf{x}} f(\mathbf{x}) = \sum_{k=1}^{4} \Delta t_k \ + \ 36\, w \sum_{i=1}^{3} \sum_{k=1}^{4} d_{ik}\,(\mathbf{x})^2\, \Delta t_k$$

Total time $T = \sum \Delta t_k$ plus the weighted squared jerk of all joints (w = smoothness weight).

Constraints

Equality constraints, $h(\mathbf{x}) = 0$ (for $i = 1,2,3$):

	Constraint	Meaning
h_1	$q_i(0) - q_i^{\text{start}} = 0, \quad q_i(T) - q_i^{\text{goal}} = 0$	Start and end at the given poses
h_2	$\dot{q}_i(0) = 0, \quad \dot{q}_i(T) = 0$	Start and end at rest
h_3	$q_{ik}(\Delta t_k) - q_i^{(k)} = 0, \quad q_{i,k+1}(0) - q_i^{(k)} = 0$	Pass through each waypoint
h_4	$\dot{q}_{ik}(\Delta t_k) - \dot{q}_{i,k+1}(0) = 0, \quad \ddot{q}_{ik}(\Delta t_k) - \ddot{q}_{i,k+1}(0) = 0$	Smooth at waypoints $k = 1,2,3$

Inequality constraints, $g(\mathbf{x}) \leq 0$ (for $i = 1,2,3$ and all $t \in [0, T]$):

	Constraint	Meaning
g_1	$\Delta t_{\min} - \Delta t_k \leq 0$	Segment durations stay positive
g_2	$q_i(t) - q_{i,\max} \leq 0, \quad q_{i,\min} - q_i(t) \leq 0$	Joint angle limits
g_3	$\pm\dot{q}_i(t) - v_{i,\max} \leq 0$	Joint velocity limit
g_4	$\pm\ddot{q}_i(t) - a_{i,\max} \leq 0$	Joint acceleration limit
g_5	$\pm u_i(t) - u_{i,\max} \leq 0$	Joint torque limit